

Lab 11 Report

Mini Project 02: Fully-Differential Folded Cascode OTA

1. Part 1: gm/ID Design Charts

1.1 Design Charts (NMOS & PMOS, $L = \min, 1\mu\text{m} : 1\mu\text{m} : 5\mu\text{m}$)

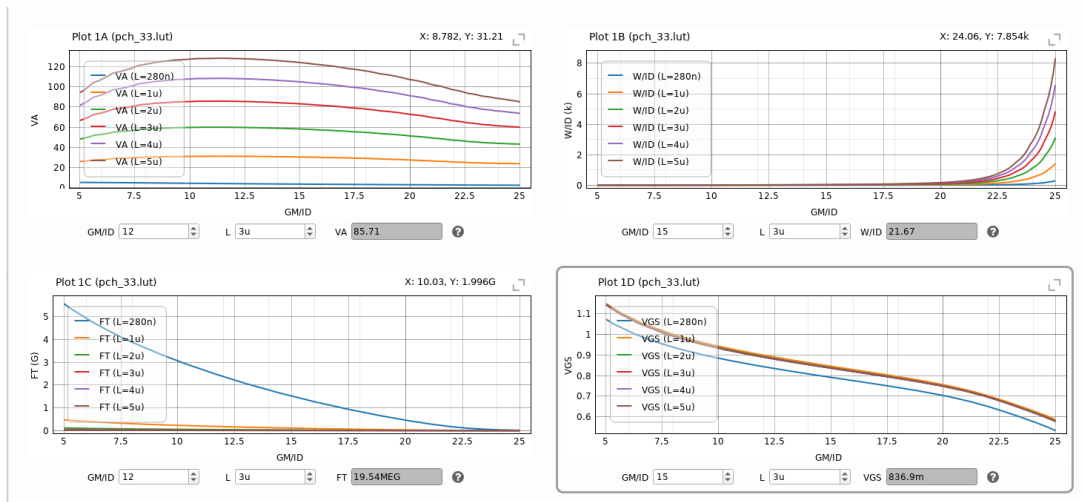


Figure 1: Design Charts for PMOS

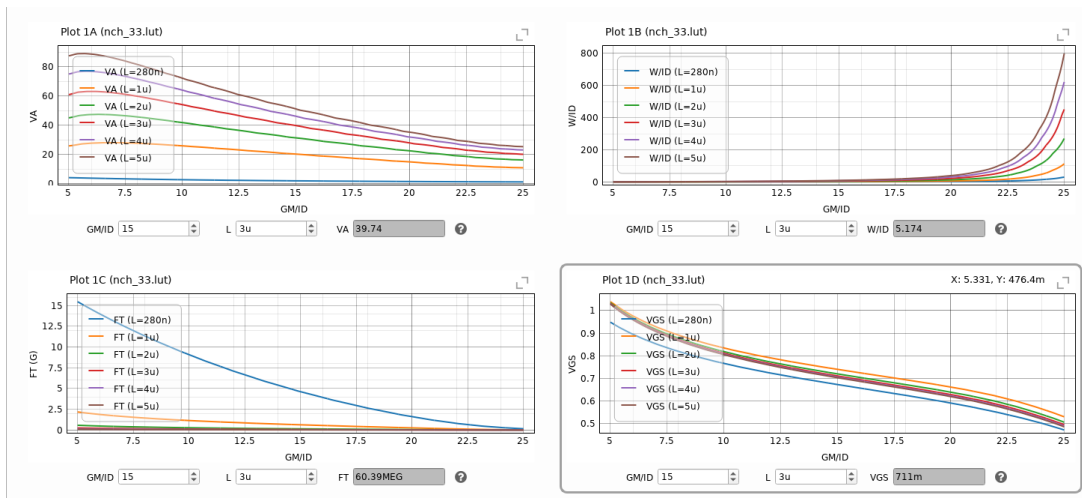


Figure 2: Design Charts for NMOS

2. Part 2: OTA Design

2.1 Specifications

Parameter	
Technology	ADT45nm
Supply voltage V_{DD}	2.5 V
Closed loop gain	2
Phase margin	$\geq 70^\circ$
CM input range – low	≤ 0 V
CM input range – high	≥ 1 V
Differential output swing	1.2 V _{pk-to-pk}
Load C_L	500 fF
DC Loop gain	60 dB
CL settling time for 1% error	100 ns

Table 1: Target Folded Cascode OTA specifications

2.2 Input Pair Type Justification

Why PMOS input stage?

From the CMIR spec ($\text{CMIR}_{\text{low}} \leq 0\text{V}$), a PMOS input stage is required to allow the input common-mode to reach ground while keeping the input devices in saturation.

2.3 Hand Analysis: Design Procedure

Hand Analysis: Settling Time to GBW Requirement

1. Time Constant (τ) Calculation:

For a first-order system, the step response error decays exponentially. To achieve a 1% settling time ($t_{\text{settle}} = 100$ ns), the error must be ≤ 0.01 :

$$\begin{aligned}
 e^{-\frac{t_{\text{settle}}}{\tau}} &= 0.01 \\
 -\frac{100 \text{ ns}}{\tau} &= \ln(0.01) \\
 \tau &= \frac{100 \text{ ns}}{\ln(100)} \approx 21.71 \text{ ns}
 \end{aligned}$$

2. Closed-Loop Bandwidth (BW_{CL}):

The time constant defines the closed-loop angular bandwidth (ω_{CL}):

$$\omega_{CL} = \frac{1}{\tau} = \frac{1}{21.71 \text{ ns}} \approx 46.05 \text{ Mrad/s}$$

$$BW_{CL} = \frac{\omega_{CL}}{2\pi} = \frac{46.05 \text{ Mrad/s}}{2\pi} \approx 7.33 \text{ MHz}$$

3. Required Open-Loop Gain-Bandwidth Product (GBW):

The closed-loop bandwidth is determined by the open-loop GBW and the feedback factor (β). Assuming $\beta = 1/3$ due to the capacitive voltage divider effect formed by C_f , C_s , and the OTA input capacitance (C_{in}):

$$\beta = \frac{C_F}{C_F + C_{in} + C_s}$$

$$BW_{CL} = \beta \times GBW$$

$$GBW = \frac{BW_{CL}}{\beta}$$

$$GBW = \frac{7.33 \text{ MHz}}{1/3} \approx 21.99 \text{ MHz}$$

Conclusion: To safely meet the 100 ns settling time specification, the OTA must be designed to achieve an open-loop GBW of approximately **22 MHz**.

Input Pair Design

In order to increase the GBW as much as possible and minimize input-referred noise, I would bias the input pair in the MI i.e $\frac{g_m}{I_D} = 17$

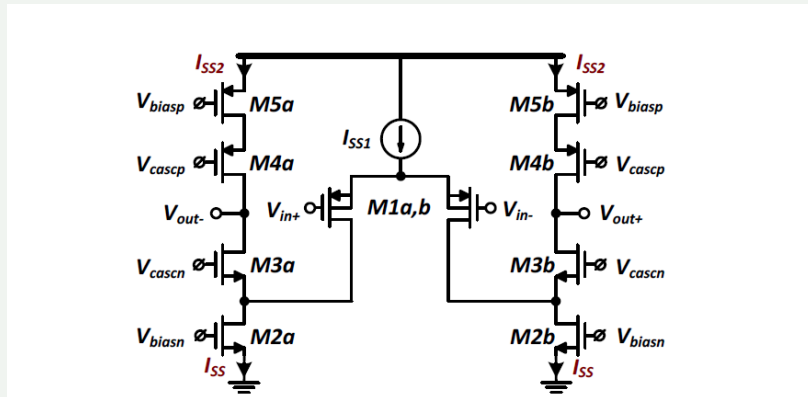


Figure 3: Folded Cascode Schematic

$$V_{nin}^2(f) = \frac{8kT\gamma}{g_{m1}} \left(1 + \frac{g_{m2}}{g_{m1}} + \frac{g_{m5}}{g_{m1}} \right) = \frac{8kT\gamma\alpha}{g_{m1}}$$

where γ is the noise coefficient ($\approx 2/3$ for long-channel devices and 1 for short channel) I will use $I_D = 10\mu A$ in the input pair to maximise g_m

4. Loading Effect & Total Capacitance ($C_{L,\text{eff}}$):

The effective load capacitance must account for the external load ($C_{load} = 500\text{ fF}$), the loading effect of the feedback network, and the OTA's parasitic output capacitance (assuming $C_{out} \approx 27\text{ fF}$):

$$C_{L,\text{eff}} = C_{load} + (1 - \beta)C_f + C_{out}$$

$$C_{L,\text{eff}} \approx 500\text{ fF} + \left(1 - \frac{1}{3}\right)(1\text{ pF}) + 27\text{ fF}$$

$$C_{L,\text{eff}} \approx 0.5\text{ pF} + 0.667\text{ pF} + 0.027\text{ pF} \approx 1.194\text{ pF}$$

5. Required Transconductance ($g_{m1,2}$):

Using the required open-loop GBW ($\omega_{GBW} = 2\pi \times 21.99\text{ MHz} \approx 138.17\text{ Mrad/s}$):

$$g_{m1,2} \geq C_{L,\text{eff}} \times \omega_{GBW}$$

$$g_{m1,2} \geq 1.194\text{ pF} \times 138.17\text{ Mrad/s} \approx 165\text{ }\mu S$$

Conclusion: To robustly meet the 100 ns settling time specification while accounting for feedback loading and parasitics, the minimum required transconductance is approximately **165 μS** . Since there is no requirement for input pair except giving high g_m for GBW. I would assume $L=310\text{ n}$ to decrease input capacitance.

The screenshot shows a configuration window for an input pair. The parameters are set as follows:

- ID: 10u
- gm/ID: 17
- L: 310n
- VDS: 800m
- VSB: 0
- Stack: 1

Below the parameters is a table titled "Results:" with the following data:

	Name	TT-27.0
4	W	24.88u
5	VDB	800m
6	VGB	773.5m
7	VGS	773.5m
8	VDS	800m

Figure 4: Input Pair

Designing the BiasN transistors

Given $g_m = 169.3\text{ }\mu S$. and from the loop gain requirement. I would need $A_{OL} > 3000$ and consequently $R_{out} > 17.72\text{ M}\Omega$.

Take notice that G_m is not exactly equal to $g_{m1,2}$ as some current leaks in the r_o of the bias transistors as a result of current divider. I would use split ratio of 2 (CS carries double the current of the CG branch) to improve the amplifier DC gain, bandwidth, and noise according to the paper*.

$$G_m = g_{m1,2} \frac{G_1}{G_1 + G_2} = g_m \frac{g_{m3}}{g_{m3} + \frac{5}{2r_o}} = \alpha g_m$$

(Assuming $r_{o1} = 3/2r_{o2}$) From the noise equation. We find that M2 is a primary contributor. I would use gm/ID of 10 (SI). also, I would try to increase its r_o to increase α

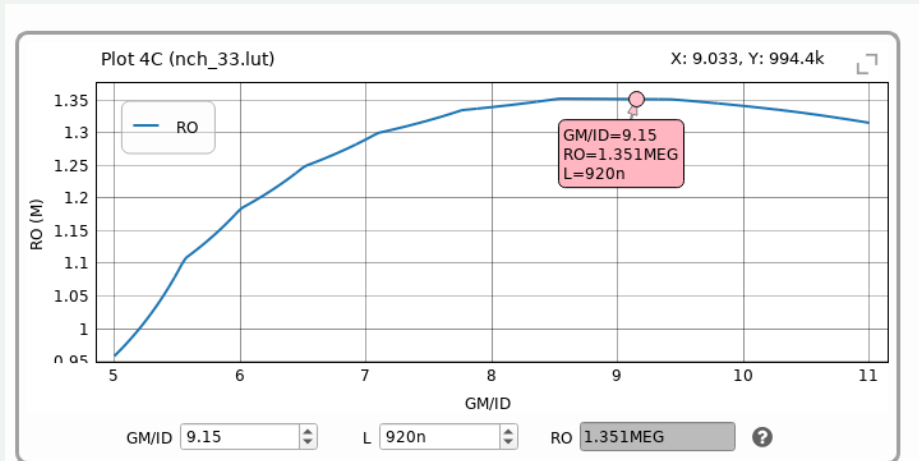


Figure 5: r_o vs $\frac{g_m}{I_D}$

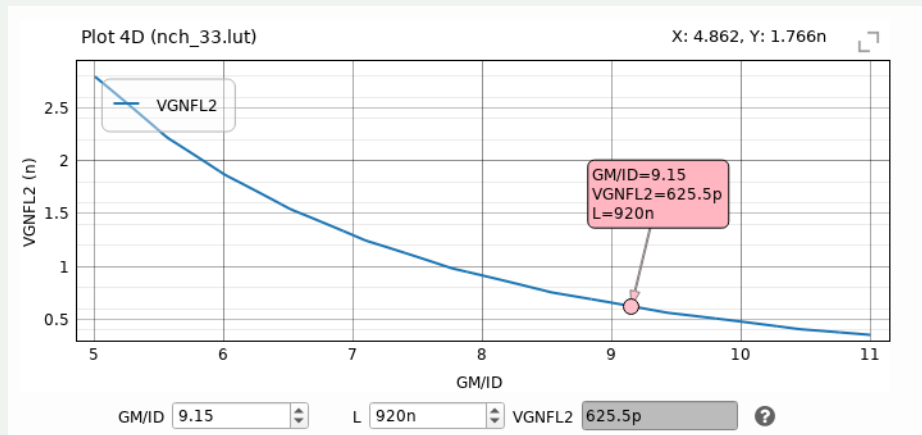


Figure 6: gate-referred flicker noise voltage spectral density at 1 Hz vs $\frac{g_m}{I_D}$

* Omran, H. (2019). Optimum split ratio for folded cascode OTA bias current: A qualitative and quantitative study. *2019 31st International Conference on Microelectronics (ICM)*, 26–29. <https://doi.org/10.1109/ICM48031.2019.9021755>

ID ?
 gm/ID ?
 L ?
 VDS ?
 VSB ?
 Stack ?

Results:

	Name	TT-27.0
3	L	920n
4	W	5.16u
5	VDB	500m
6	VGB	858.5m
7	VGS	858.5m

Figure 7: Sizing of BiasN Transistors

Step 3: Designing the current mirror

Frequency ?

ID ?
 gm/ID ?
 L ?
 VDS ?
 VSB ?
 Stack ?

Results:

	Name	TT-27.0
3	L	1u
4	W	9.64u

Figure 8: Current Mirror Sizing

Step 3: Designing the CASCN transistors

To achieve $R_{out} = 17.72M\Omega$ $R_{down} = g_{m,casc}r_{o,casn}(r_{o,biasn}/r_{o,input}) \approx 20M\Omega$. I will need $gm/gds = 70$

Results:

	Name	TT-27.0
3	L	390n
4	W	1.6u

Figure 9: CASCN sizing

Step 4: CASCP Transistors

I would use gm/gds of 70 to achieve R_{out} spec

Results:

	Name	TT-27.0
3	L	360n
4	W	4.4u

Figure 10: Sizing of CASCP transistors

Determining V_{cascn} & V_{cascp} & V_{ref}

$$V_{CASCN} = V_{GScascN} + V_{biasN}^* = 1.11 \text{ V}$$

$$V_{CASCp} = V_{DD} - V_{SGcascP} - V_{biasP}^* = 1.26 \text{ V}$$

$$V_{out,min} = V_{CASCN} + V_{biasN}^* = 393.5 \text{ mV}$$

$$V_{out,max} = V_{DD} - V_{SG} - V^* = 1.4765 \text{ V}$$

$$V_{ref} \approx 935 \text{ mV}$$

Note: I used slightly higher voltage for cascn and lower for cascp to keep the NMOS and PMOS in saturation.

2.4 Designing the CMFB loop and the biasing circuit

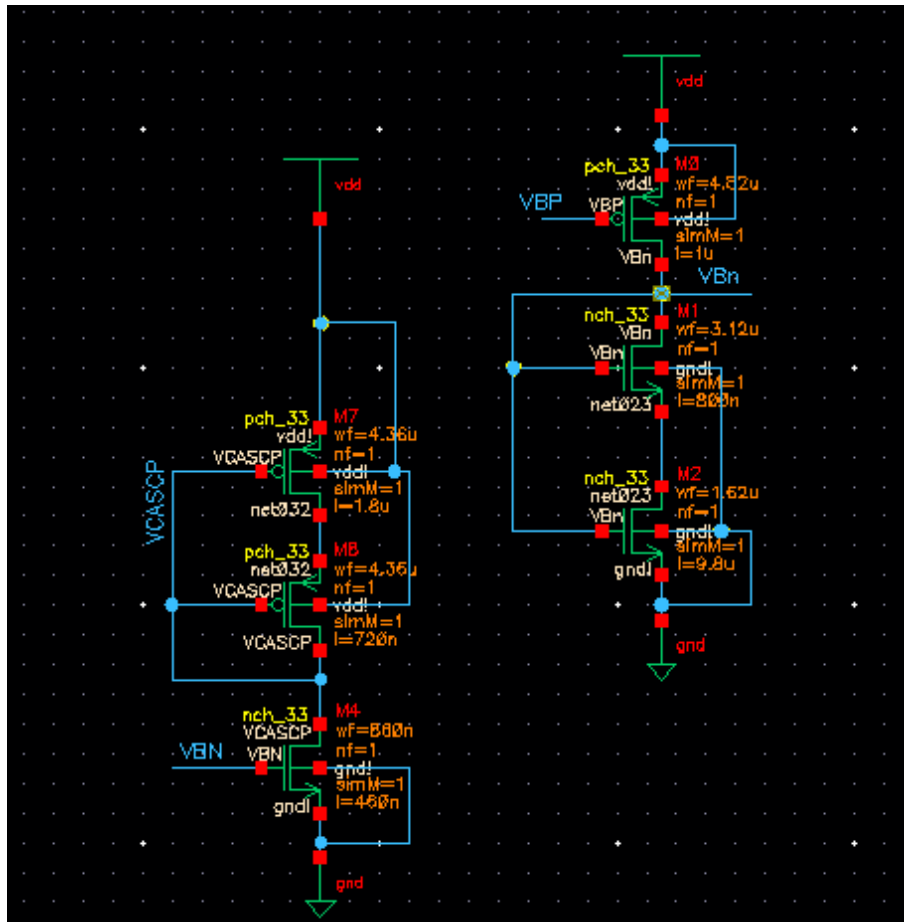


Figure 11: Biasing circuit for CASCn and CASCp

the NMOS biasing allows $2.5 \mu\text{A}$ while the PMOS allows $5 \mu\text{A}$. As for the sizing of M4, they are sized as the current source driving the folded cascode with scaling W . As for M1, M8 they are sized the same as CASCn and CASCp transistors, respectively, while scaling W to provide the same bias point.

As for M2, after fine-tuning, I got $W=1.62\mu$. To get L , I increased above the required $1.11V$

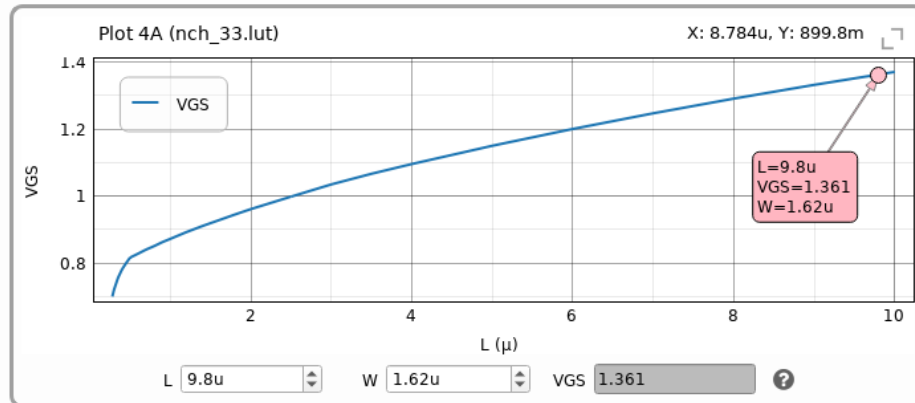


Figure 12: VGS vs L

As for M7, after fine-tuning, Using the same W of M6. To get L , I decreased below the required $1.26V$ as $V_{CASCP} = V_{DD} - V_{SG7}$

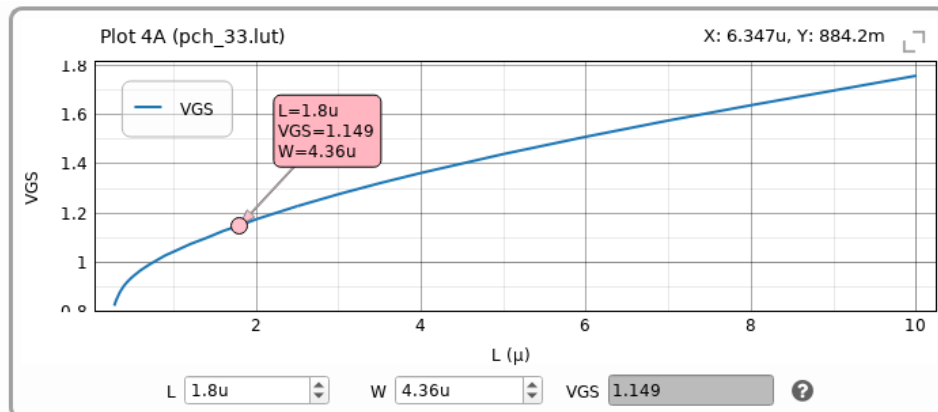


Figure 13: VGS vs L

2.4.1 Designing the CMFB loop

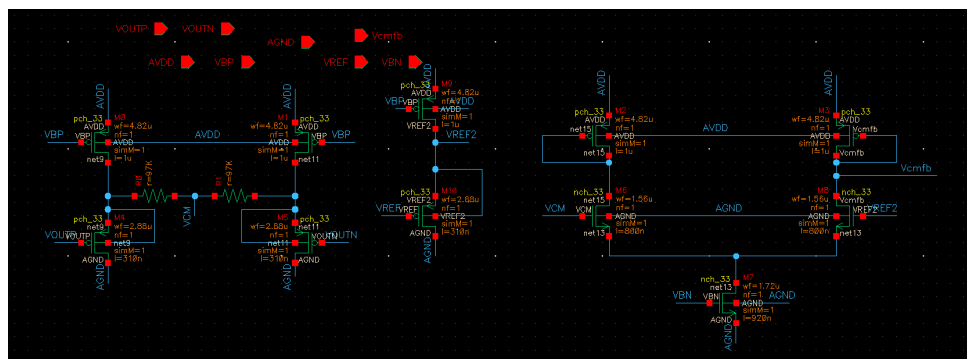


Figure 14: The CMFB schematic

The basic idea of the CMFB loop is to use common drain to avoid the loading effect and then use matched resistors to get V_{OCM} . Also, We need another common drain for the V_{ref} as well. Then we compare the two voltages with low gain amplifier to avoid stability issues and get V_{cmfb} . As for M_0, M_1, M_9 . They represent the current source for the common drain. M_7 is biased the same as the biasN transistor. As for the other transistors. They were fine tuned through trial and error.

3. Part 3: Open-Loop OTA Simulation (Behavioral CMFB)

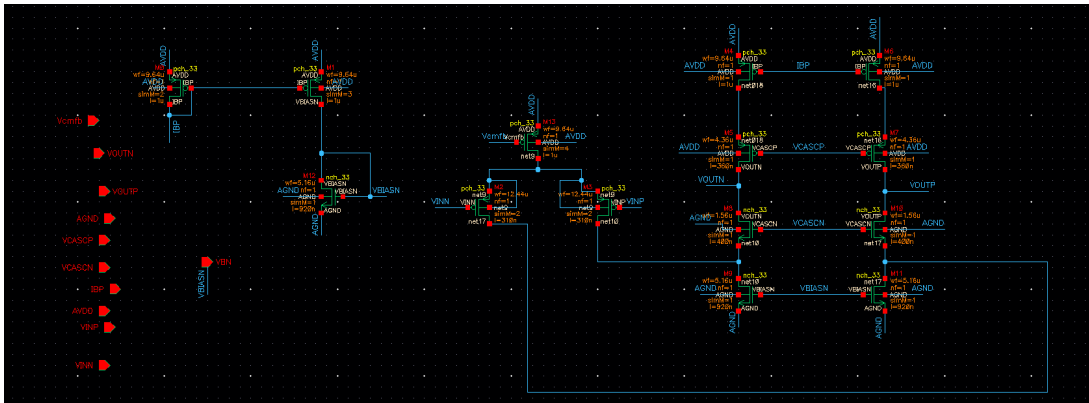


Figure 15: Folded Fully Differential Cascode OTA

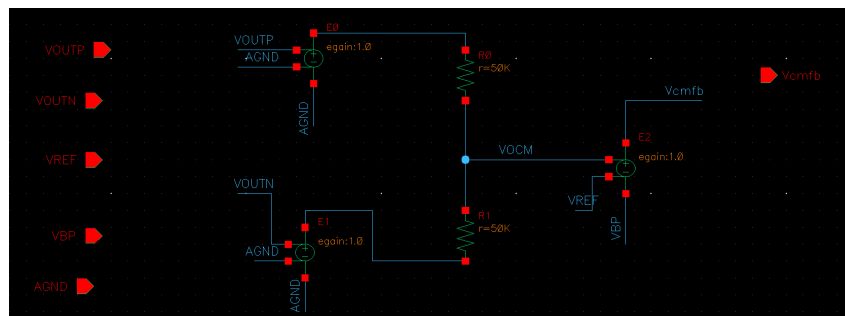


Figure 16: Behavioral model of CMFB

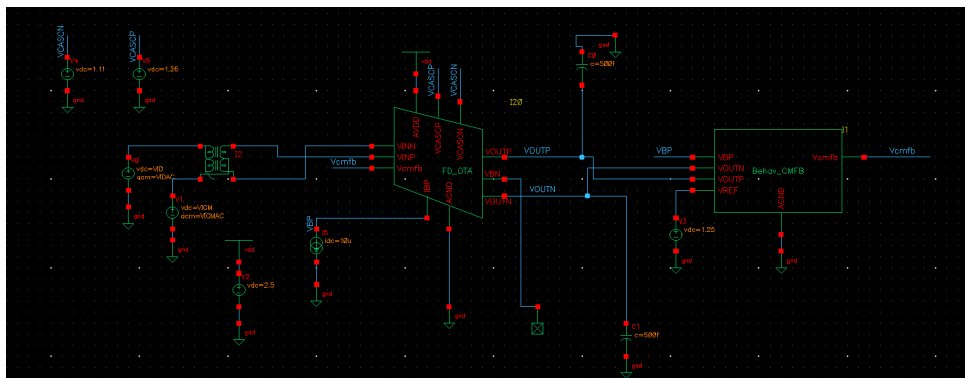


Figure 17: Behavioral TestBench

Transistor Operating Points (Open-Loop CMFB)										
Instance	Type	Region	Id	Vgs	Vds	Vdsat	gm	vth	gds	gm/Id
I20.M0.m0	pch_33	sat	-10 uA	-942.852 mV	-942.852 mV	-155.355 mV	100.444 uS	-783.92 mV	229.3 nS	10.0444 1/V
I20.M1.m0	pch_33	sat	-15.1959 uA	-942.852 mV	-1.64311 V	-155.356 mV	152.489 uS	-783.918 mV	234.62 nS	10.0349 1/V
I20.M2.m0	pch_33	sat	-9.82155 uA	-775.103 mV	-1.01915 V	-91.5308 mV	168.641 uS	-751.568 mV	2.22658 uS	17.1705 1/V
I20.M3.m0	pch_33	sat	-9.82155 uA	-775.103 mV	-1.01915 V	-91.5308 mV	168.641 uS	-751.568 mV	2.22658 uS	17.1705 1/V
I20.M4.m0	pch_33	sat	-4.86294 uA	-942.852 mV	-265.801 mV	-155.353 mV	48.5807 uS	-783.922 mV	638.318 nS	9.99 1/V
I20.M5.m0	pch_33	sat	-4.86306 uA	-974.199 mV	-981.833 mV	-149.905 mV	58.6747 uS	-863.681 mV	623.322 nS	12.0654 1/V
I20.M6.m0	pch_33	sat	-4.86294 uA	-942.852 mV	-265.801 mV	-155.353 mV	48.5807 uS	-783.922 mV	638.318 nS	9.99 1/V
I20.M7.m0	pch_33	sat	-4.86306 uA	-974.199 mV	-981.833 mV	-149.905 mV	58.6747 uS	-863.681 mV	623.322 nS	12.0654 1/V
I20.M8.m0	nch_33	sat	4.86347 uA	854.05 mV	996.417 mV	140.459 mV	59.2187 uS	761.93 mV	693.2 nS	12.1762 1/V
I20.M9.m0	nch_33	sat	14.6843 uA	856.894 mV	255.95 mV	187.354 mV	133.36 uS	689.767 mV	3.03039 uS	9.08179 1/V
I20.M10.m0	nch_33	sat	4.86347 uA	854.05 mV	996.417 mV	140.459 mV	59.2187 uS	761.93 mV	693.2 nS	12.1762 1/V
I20.M11.m0	nch_33	sat	14.6843 uA	856.894 mV	255.95 mV	187.354 mV	133.36 uS	689.767 mV	3.03039 uS	9.08179 1/V
I20.M12.m0	nch_33	sat	15.1959 uA	856.894 mV	856.894 mV	187.357 mV	139.147 uS	689.763 mV	493.787 nS	9.15691 1/V
I20.M13.m0	pch_33	sat	-19.6428 uA	-940.486 mV	-1.2249 V	-153.74 mV	159.194 uS	-783.919 mV	372.388 nS	10.1408 1/V

Figure 18: DC operating points at $V_{ICM} = 0.5$

DC Operating Point Questions

Q: What is the CM level at the OTA output?

1.252V

Q: What are the differential input and output voltages of the error amplifier? What is the relation between them?

$V_{in+} = 1.252V$.

$V_{in-} = 1.25V$.

$V_{out+} = 1.56V$.

$V_{out-} = 1.557V$.

They are nearly the same.

3.1 Differential Small-Signal AC Analysis

Test	Output	Nominal
FoldedCascode:TB:1	A0	3.17k
FoldedCascode:TB:1	BW	15.67k
FoldedCascode:TB:1	UGF	49.66M
FoldedCascode:TB:1	GBW	49.78M
FoldedCascode:TB:1	PM	83.32

Figure 19: Small signal parameters

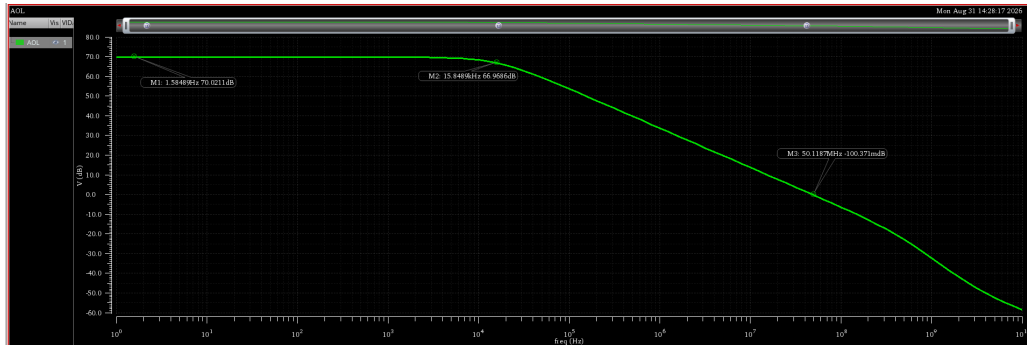


Figure 20: Differential small signal gain vs frequency

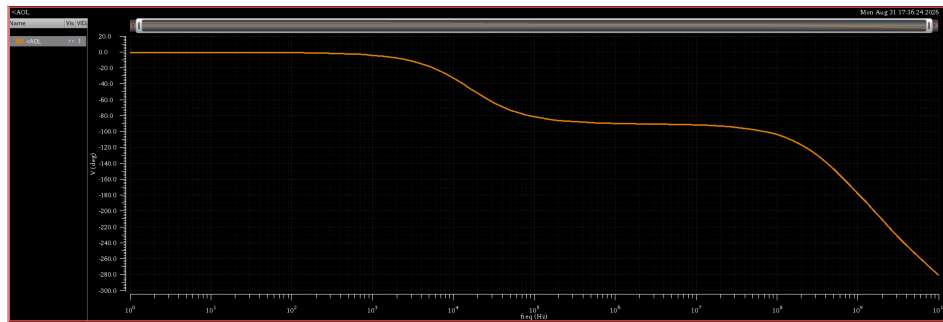


Figure 21: Phase vs frequency

1. Effective Transconductance (G_m):

$$g_{m1,2} = 168.64 \mu\text{S}$$

$$\alpha = \frac{g_{m3} + g_{mb3}}{(g_{m3} + g_{mb3}) + g_{ds1,2} + g_{ds_bot}}$$

$$\alpha = \frac{59.22 \mu\text{S} + 16.23 \mu\text{S}}{75.45 \mu\text{S} + 2.23 \mu\text{S} + 3.03 \mu\text{S}} = \frac{75.45 \mu\text{S}}{80.71 \mu\text{S}} \approx 0.935$$

$$G_m = \alpha g_{m1,2} = 0.935 \times 168.64 \mu\text{S} \approx 157.66 \mu\text{S}$$

2. Output Resistance (R_{out}):

$$R_{down} \approx \frac{g_{m3} + g_{mb3}}{g_{ds3}(g_{ds1,2} + g_{ds_bot})} = \frac{75.45 \mu\text{S}}{0.693 \mu\text{S} \times 5.26 \mu\text{S}} \approx 20.71 \text{ M}\Omega$$

$$R_{up} \approx \frac{g_{m4} + g_{mb4}}{g_{ds4}g_{ds_top}} = \frac{48.58 \mu\text{S} + 22.96 \mu\text{S}}{0.638 \mu\text{S} \times 0.623 \mu\text{S}} \approx 179.99 \text{ M}\Omega$$

$$R_{out} = R_{down} \parallel R_{up} = 20.71 \text{ M}\Omega \parallel 179.99 \text{ M}\Omega \approx 18.57 \text{ M}\Omega$$

3. DC Gain (A_0):

$$A_0 = G_m \times R_{out}$$

$$A_0 = 157.66 \mu\text{S} \times 18.57 \text{ M}\Omega \approx 2927 \text{ V/V}$$

4. Gain-Bandwidth Product (GBW) & Bandwidth (BW):

$$GBW = \frac{G_m}{2\pi C_L} = \frac{157.66 \mu S}{2\pi \times 500 \text{ fF}} \approx 50.18 \text{ MHz}$$

$$BW = \frac{GBW}{A_0} = \frac{50.18 \text{ MHz}}{2927} \approx 17.14 \text{ kHz}$$

5. Phase Margin (PM):

5. Phase Margin (PM) from PMOS Cascode Pole:

The resistance at the PMOS cascode node (M_4/M_5) is dominated by looking up into the source of M_5 , while the capacitance is the sum of the junction and overlap capacitances of both M_4 and M_5 .

$$R_{cascP} \approx \frac{1}{g_{m5} + g_{mb5}}$$

$$R_{cascP} \approx \frac{1}{58.67 \mu S + 19.75 \mu S} = \frac{1}{78.42 \mu S} \approx 12.75 \text{ k}\Omega$$

$$C_{cascP} \approx C_{db4} + C_{gd4} + C_{gs5} + C_{sb5}$$

$$C_{cascP} \approx 5.20 \text{ fF} + 2.52 \text{ fF} + 4.56 \text{ fF} + 0.82 \text{ fF} = 13.10 \text{ fF}$$

$$f_{p,cascP} = \frac{1}{2\pi R_{cascP} C_{cascP}}$$

$$f_{p,cascP} = \frac{78.42 \mu S}{2\pi \times 13.10 \text{ fF}} \approx 952.8 \text{ MHz}$$

$$PM = 180^\circ - 90^\circ - \arctan\left(\frac{GBW}{f_{p,cascP}}\right)$$

$$PM \approx 90^\circ - \arctan\left(\frac{50.18 \text{ MHz}}{952.8 \text{ MHz}}\right)$$

$$PM \approx 90^\circ - 3.02^\circ \approx 86.98^\circ$$

Diff Gain: Hand Calc vs. Simulation

Parameter	Hand Analysis	Simulation
DC Gain (A_o) (dB)	69.3	70.0
BW (Hz)	17.14k	15.67k
GBW (Hz)	50.18M	49.78M
UGF (f_u) (Hz)	50.18M	49.66M
Phase Margin ($^\circ$)	86.98	83.32

4. Part 4: Open-Loop OTA Simulation (Actual CMFB)

Transistor Operating Points (Open-Loop Real CMFB)												
Instance	Type	Region	Id	Vgs	Vds	Vdsat	gm	vth	gds	gm/Id		
10.M0.MD	pch_33	sat	-10 uA	-942.852 mV	-942.852 mV	-150.250 mV	100.444 uS	-783.92 mV	229.3 nS	10.0444	1/V	
10.M1.MD	pch_33	sat	-15.1359 uA	-942.852 mV	-1.64311 V	-150.256 mV	152.489 uS	-783.618 mV	234.42 nS	10.0349	1/V	
10.M2.MD	pch_33	sat	-9.53261 uA	-776.407 mV	-970.413 mV	-50.013 mV	170.076 uS	-751.467 mV	2.27113 uS	17.1226	1/V	
10.M3.MD	pch_33	sat	-9.53261 uA	-776.407 mV	-970.413 mV	-50.013 mV	170.076 uS	-751.467 mV	2.27113 uS	17.1226	1/V	
10.M4.MD	pch_33	sat	-4.86171 uA	-942.852 mV	-253.859 mV	-150.253 mV	48.259 uS	-783.522 mV	653.528 nS	9.88805	1/V	
10.M5.MD	pch_33	sat	-4.86151 uA	-970.252 mV	-1.30647 V	-148.703 mV	56.0253 uS	-981.34 mV	568.659 nS	12.1364	1/V	
10.M6.MD	pch_33	sat	-4.86171 uA	-942.852 mV	-253.859 mV	-150.253 mV	48.259 uS	-783.522 mV	653.528 nS	9.88805	1/V	
10.M7.MD	pch_33	sat	4.86251 uA	-970.252 mV	-1.30647 V	-148.703 mV	56.0253 uS	-981.34 mV	568.659 nS	12.1364	1/V	
10.M8.MD	pch_33	sat	4.86173 uA	-872.252 mV	623.638 mV	142.853 mV	58.8024 uS	777.072 mV	820.004 nS	12.005	1/V	
10.M9.MD	nch_33	sat	4.86173 uA	872.252 mV	623.638 mV	142.853 mV	58.8024 uS	777.072 mV	820.004 nS	12.005	1/V	
10.M10.MD	nch_33	sat	4.86173 uA	872.252 mV	623.638 mV	142.853 mV	58.8024 uS	777.072 mV	820.004 nS	12.005	1/V	
10.M11.MD	nch_33	sat	15.1359 uA	856.894 mV	253.944 mV	187.357 mV	130.147 uS	699.763 mV	492.787 nS	9.14691	1/V	
10.M12.MD	nch_33	sat	15.1359 uA	856.894 mV	253.944 mV	187.357 mV	130.147 uS	699.763 mV	492.787 nS	9.14691	1/V	
11.M0.MD	pch_33	sat	-2.48414 uA	-942.852 mV	-759.005 mV	-150.758 mV	24.7964 uS	-783.565 mV	69.1145 nS	9.9819	1/V	
11.M1.MD	pch_33	sat	-2.48414 uA	-942.852 mV	-759.005 mV	-150.758 mV	24.7964 uS	-783.565 mV	69.1145 nS	9.9819	1/V	
11.M2.MD	pch_33	sat	-2.31023 uA	-595.298 mV	-595.298 mV	-150.607 mV	23.7987 uS	-783.364 mV	54.146 nS	10.3006	1/V	
11.M3.MD	pch_33	sat	-2.48454 uA	-941.034 mV	-941.034 mV	-154.909 mV	34.7254 uS	-783.564 mV	57.3022 nS	10.0025	1/V	
11.M4.MD	pch_33	sat	-2.48413 uA	-810.483 mV	-1.7403 V	-121.356 mV	35.1548 uS	-738.625 mV	434.744 nS	14.1518	1/V	
11.M5.MD	pch_33	sat	-2.48413 uA	-810.483 mV	-1.7403 V	-121.356 mV	35.1548 uS	-738.625 mV	434.744 nS	14.1518	1/V	
11.M6.MD	nch_33	sat	2.31023 uA	1.01782 V	842.464 mV	143.223 mV	28.6794 uS	919.38 mV	106.233 nS	12.4141	1/V	
11.M7.MD	nch_33	sat	4.77476 uA	856.894 mV	722.270 mV	186.507 mV	43.3655 uS	685.7 mV	194.285 nS	9.98651	1/V	
11.M8.MD	nch_33	sat	2.48454 uA	1.02311 V	836.118 mV	147.008 mV	29.8983 uS	919.38 mV	112.225 nS	12.1314	1/V	
11.M9.MD	nch_33	sat	-2.48377 uA	-942.852 mV	-754.012 mV	-150.718 mV	24.7926 uS	-783.593 mV	69.4635 nS	9.97642	1/V	
11.M10.MD	nch_33	sat	-2.48377 uA	-942.852 mV	-754.012 mV	-150.718 mV	24.7926 uS	-783.593 mV	69.4635 nS	9.97642	1/V	
M1.MD	nch_33	sat	2.515 uA	852.592 mV	852.592 mV	108.745 mV	38.2205 uS	805.071 mV	130.938 nS	15.197	1/V	
M2.MD	nch_33	triode	2.515 uA	1.17623 V	325.084 mV	468.295 mV	7.65914 uS	688.827 mV	2.54154 uS	3.04062	1/V	
M4.MD	nch_33	sat	5.09886 uA	856.894 mV	1.28581 V	196.059 mV	44.8702 uS	695.946 mV	383.395 nS	8.80005	1/V	
M7.MD	pch_33	triode	-5.09886 uA	-1.23419 V	147.037 mV	-352.371 mV	14.1228 uS	-775.902 mV	20.6526 uS	2.76976	1/V	
RE.MD	pch_33	sat	-5.09886 uA	-1.09525 V	-1.09525 V	-200.665 mV	42.373 uS	-952.962 mV	147.949 nS	8.31029	1/V	

Figure 22: DC Operating Points

Test	Output	Nominal
FoldedCascode:TB_REAL:1	A0	3.579k
FoldedCascode:TB_REAL:1	BW	14.23k
FoldedCascode:TB_REAL:1	UGF	50.77M
FoldedCascode:TB_REAL:1	GBW	50.92M
FoldedCascode:TB_REAL:1	PM	83.02

Figure 23: Differential Small Signal Parameters

Actual CMFB Questions

Q: What is the CM level at the OTA output?
929.6mV

Q: What are the differential input and output voltages of the error amplifier? What is the relation between them?

$$V_{in+} = 1.745V.$$

$$V_{in-} = 1.74V.$$

$$V_{out+} = 1.565V.$$

$$V_{out-} = 1.558V.$$

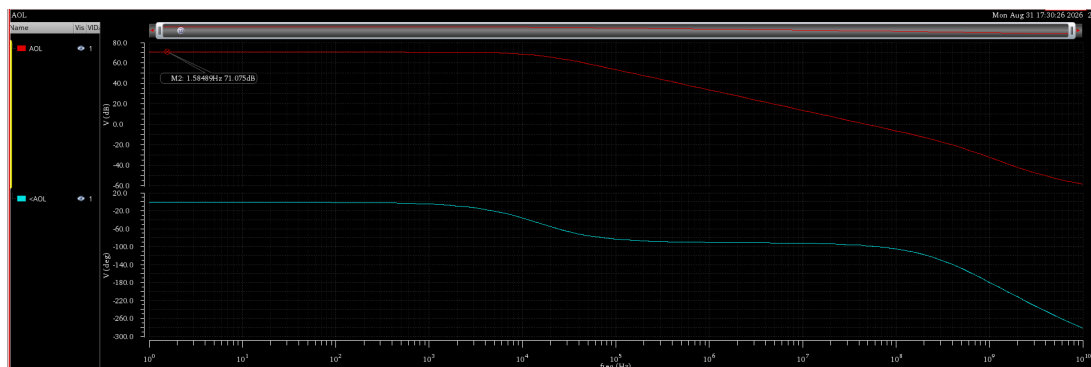


Figure 24: Open Loop DC gain Magnitude in dB and phase vs frequency

5. Part 5: Closed Loop Simulation (AC and STB Analysis)

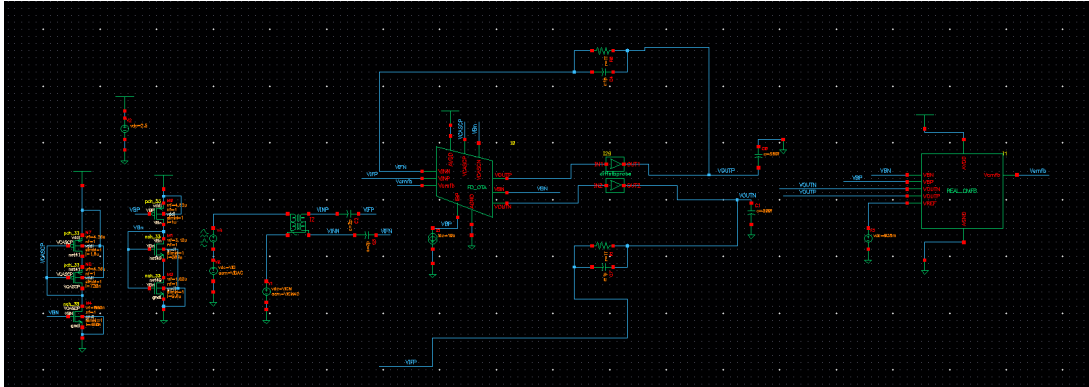


Figure 25: Closed Loop TestBench

Instance	Type	Region	Id	Vgs	Vds	Vdsat	gm	Vth	gds	gm/Id
I0.M0.m0	pch_33	sat	-10 uA	-942.852 mV	-942.852 mV	-155.355 mV	100.444 uS	-783.92 mV	229.3 nS	10.0444 1/V
I0.M1.m0	pch_33	sat	-15.1959 uA	-942.852 mV	-1.64311 V	-155.356 mV	152.489 uS	-783.918 mV	234.62 nS	10.0349 1/V
I0.M2.m0	pch_33	sat	-9.93407 uA	-770.992 mV	-1.39306 V	-90.9719 mV	170.673 uS	-748.509 mV	2.117 uS	17.1806 1/V
I0.M3.m0	pch_33	sat	-9.93407 uA	-770.992 mV	-1.39306 V	-90.9719 mV	170.673 uS	-748.509 mV	2.117 uS	17.1806 1/V
I0.M4.m0	pch_33	sat	-4.86173 uA	-942.852 mV	-263.932 mV	-155.353 mV	48.5594 uS	-783.922 mV	653.645 nS	9.98809 1/V
I0.M5.m0	pch_33	sat	-4.86173 uA	-970.259 mV	-1.30802 V	-148.676 mV	59.013 uS	-861.343 mV	568.305 nS	12.1383 1/V
I0.M6.m0	pch_33	sat	-4.86173 uA	-942.852 mV	-263.932 mV	-155.353 mV	48.5594 uS	-783.922 mV	653.645 nS	9.98809 1/V
I0.M7.m0	pch_33	sat	-4.86173 uA	-970.259 mV	-1.30802 V	-148.676 mV	59.013 uS	-861.343 mV	568.305 nS	12.1383 1/V
I0.M8.m0	nch_33	sat	4.86173 uA	872.299 mV	622.072 mV	142.812 mV	58.8001 uS	777.075 mV	820.942 nS	12.0945 1/V
I0.M9.m0	nch_33	sat	14.7945 uA	856.894 mV	305.977 mV	187.354 mV	135.295 uS	689.766 mV	1.64752 uS	9.14495 1/V
I0.M10.m0	nch_33	sat	4.86173 uA	872.299 mV	622.072 mV	142.812 mV	58.8001 uS	777.075 mV	820.942 nS	12.0945 1/V
I0.M11.m0	nch_33	sat	14.7945 uA	856.894 mV	305.977 mV	187.354 mV	135.295 uS	689.766 mV	1.64752 uS	9.14495 1/V
I0.M12.m0	nch_33	sat	15.1959 uA	856.894 mV	856.894 mV	187.357 mV	139.147 uS	689.763 mV	493.787 nS	9.15691 1/V
I0.M13.m0	nch_33	sat	-19.8656 uA	-942.526 mV	-800.959 mV	-155.131 mV	199.828 uS	-783.92 mV	516.424 nS	10.059 1/V
I1.M0.m0	pch_33	sat	-2.48424 uA	-942.852 mV	-761.465 mV	-155.758 mV	24.7975 uS	-783.395 mV	68.0068 nS	9.9819 1/V
I1.M1.m0	pch_33	sat	-2.48424 uA	-942.852 mV	-761.465 mV	-155.758 mV	24.7975 uS	-783.395 mV	68.0068 nS	9.9819 1/V
I1.M2.m0	pch_33	sat	-2.28743 uA	-934.299 mV	-934.299 mV	-149.96 mV	23.6564 uS	-783.394 mV	53.6739 nS	10.3419 1/V
I1.M3.m0	pch_33	sat	-2.48745 uA	-942.526 mV	-942.526 mV	-155.536 mV	24.8605 uS	-783.394 mV	57.8 nS	9.99436 1/V
I1.M4.m0	pch_33	sat	-2.48426 uA	-810.486 mV	-1.73854 V	-121.363 mV	35.1556 uS	-738.637 mV	434.831 nS	14.1513 1/V
I1.M5.m0	pch_33	sat	-2.48426 uA	-810.486 mV	-1.73854 V	-121.363 mV	35.1556 uS	-738.637 mV	434.831 nS	14.1513 1/V
I1.M6.m0	nch_33	sat	2.28767 uA	1.01685 V	844.02 mV	142.653 mV	28.4976 uS	919.214 mV	105.321 nS	12.457 1/V
I1.M7.m0	nch_33	sat	4.77465 uA	856.894 mV	721.68 mV	186.527 mV	43.3946 uS	685.7 mV	194.375 nS	9.08854 1/V
I1.M8.m0	nch_33	sat	2.48745 uA	1.02371 V	835.794 mV	147.553 mV	30.0758 uS	919.215 mV	113.115 nS	12.091 1/V
I1.M9.m0	pch_33	sat	-2.48377 uA	-942.852 mV	-754.612 mV	-155.758 mV	24.7928 uS	-783.395 mV	68.4835 nS	9.98191 1/V
I1.M10.m0	pch_33	sat	-2.48377 uA	-810.388 mV	-1.74539 V	-121.334 mV	35.1532 uS	-738.581 mV	434.461 nS	14.1532 1/V
M0.m0	pch_33	sat	-2.515 uA	-942.852 mV	-1.32172 V	-155.76 mV	25.0907 uS	-783.393 mV	45.7302 nS	9.97642 1/V
M1.m0	nch_33	sat	2.515 uA	852.592 mV	852.592 mV	108.745 mV	38.2205 uS	805.071 mV	130.538 nS	15.197 1/V
M2.m0	nch_33	triode	2.515 uA	1.17828 V	325.684 mV	404.296 mV	7.65974 uS	648.827 mV	2.54158 uS	3.04562 1/V
M4.m0	nch_33	sat	5.09886 uA	856.894 mV	1.26581 V	196.099 mV	44.8702 uS	686.946 mV	383.396 nS	8.80005 1/V
M7.m0	pch_33	triode	-5.09886 uA	-1.23419 V	-167.937 mV	-352.371 mV	14.1226 uS	-775.902 mV	20.6526 uS	7.66976 1/V
M8.m0	pch_33	sat	-5.09886 uA	-1.06625 V	-1.06625 V	-200.665 mV	42.373 uS	-862.982 mV	147.949 nS	8.31029 1/V

Figure 26: Closed Loop DC Operating points

Closed-Loop DC OP Questions

Q: What is the CM level at the OTA output? Why?

928mV

What are the differential input and output voltages of the error amplifier?
What is the relation between them?

$$V_{IN,diff} = 6 \text{ mV}$$

$$V_{OUT,diff} = 9 \text{ mV}$$

$$\text{Ratio} = \frac{9}{6} = 1.5$$

They are related to the small signal gain of the amplifier.

5.1 Differential Closed-Loop Response

Test	Output	Nominal
FoldedCascode:TB_CL:1	ACL	1.998
FoldedCascode:TB_CL:1	BWCL	9.038M
FoldedCascode:TB_CL:1	GBWCL	18.06M

Figure 27: Closed Loop Results

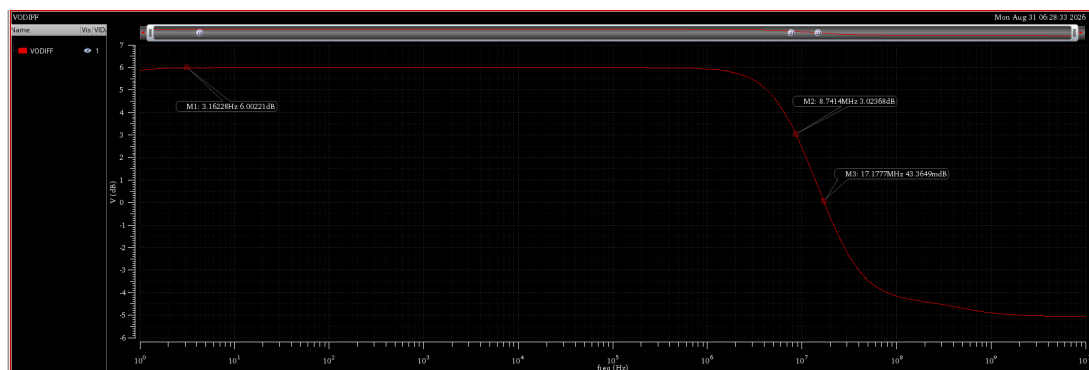


Figure 28: Closed loop gain vs frequency

5.2 Stability (STB Analysis)

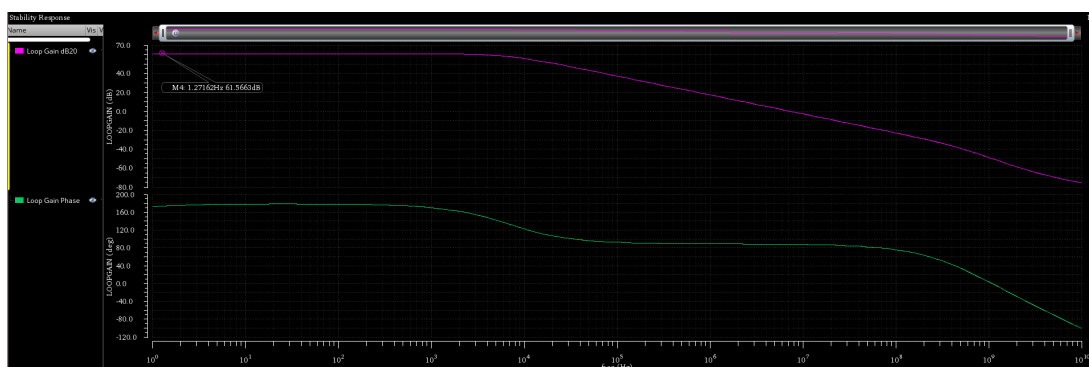


Figure 29: Differential Loop Gain and Phase vs frequency

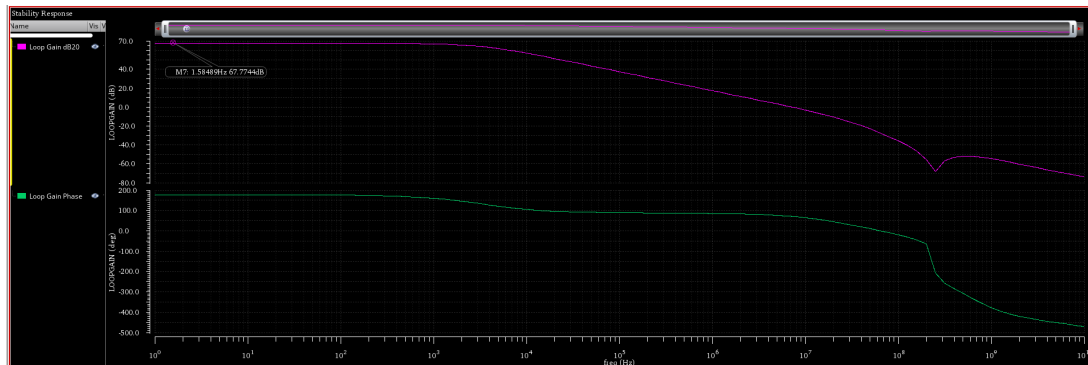


Figure 30: Common Mode Loop Gain and Phase vs frequency

Test	Output	Nominal
FoldedCascode:TB_CL:1	PM_CM	72.06
FoldedCascode:TB_CL:1	PMF_CM	7.318M

Figure 31: Common-Mode Loop Stability Parameters

Test	Output	Nominal
FoldedCascode:TB_CL:1	PMF_DIFF	7.236M
FoldedCascode:TB_CL:1	PM_DIFF	89.09

Figure 32: Differential Loop Stability Parameters

Figure 33: Comparison of Phase Margin (PM) and Phase Margin Frequency (PMF) for Closed-Loop Configurations. P.S: Ignore the PMF_Outputs. They are not accurate. i.e: don't quantify the GBWCL

STB Analysis Questions

Q: Compare GBW and PM of diff and CM loops. Comment.

Differential has less GBW but higher PM compared to CM, Making CM less stable

Q: Compare DC LG and GBW of the diff loop with those obtained from open-loop simulation. Comment.

The AOL gain in the open-loop simulation was equal to 3579, which gives a LG third that is equal to 1193, which is close to the value we get. It is a bit different due to the loading effect of the feedback that we don't see in open loop, we tried to account for it though in the open-loop simulation when designing the amplifier. Due to the lower gain we also get lower GBW

6. Part 6: Closed Loop Simulation (Transient Analysis)

6.1 Differential Input Pulse

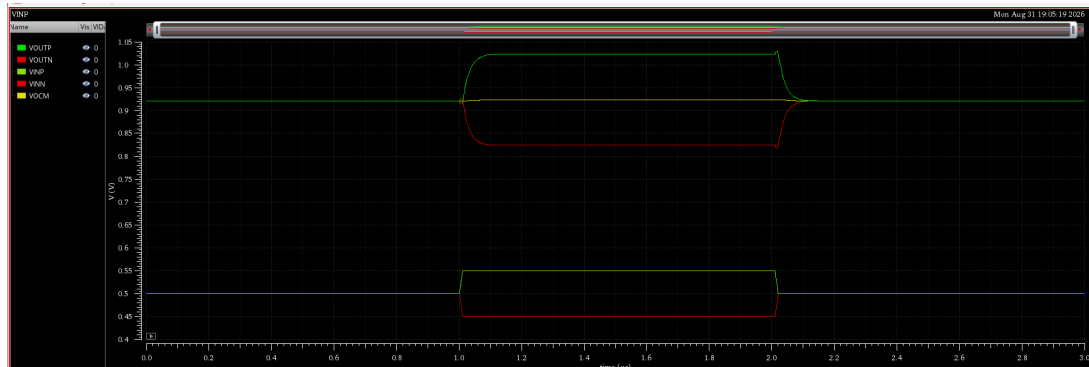


Figure 34: Transient signals at V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , and V_{OCM} .

Differential Pulse Analysis

Q: Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

There is a little bit of barely visible ringing in CM (as shown below) due to its lower PM and none in Differential. The spikes seen are caused by the capacitors in the circuit not allowing sudden changes. Both have adequate PM that is within spec and doesn't distort the signal too much.

Q: Calculate the 1% settling time and compare it to the required specification. If not satisfied, what design changes could be a solution?

FoldedCascode:TB_CL:1	Settletime	102.7n	range 100n 1...	near
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Figure 35: Settling Time

6.2 Common-Mode Input Pulse



Figure 36: Transient signals responding to CM pulse

6.3 Output Swing

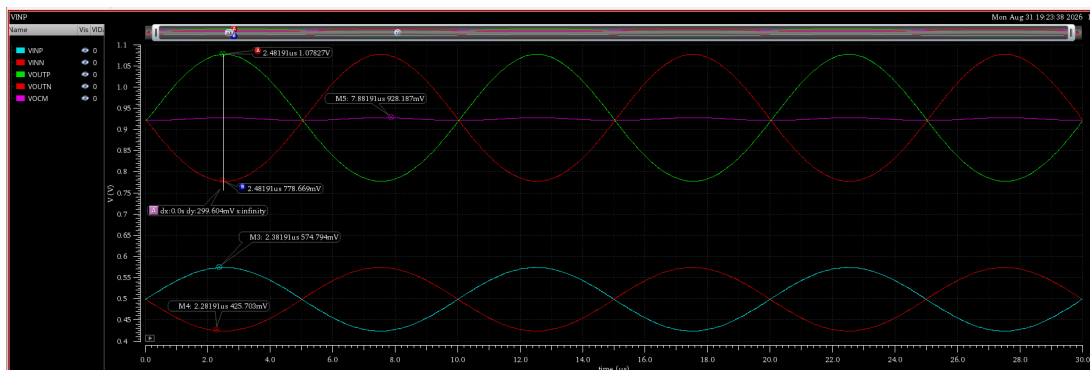


Figure 37: V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , V_{OCM} with 100kHz sine wave

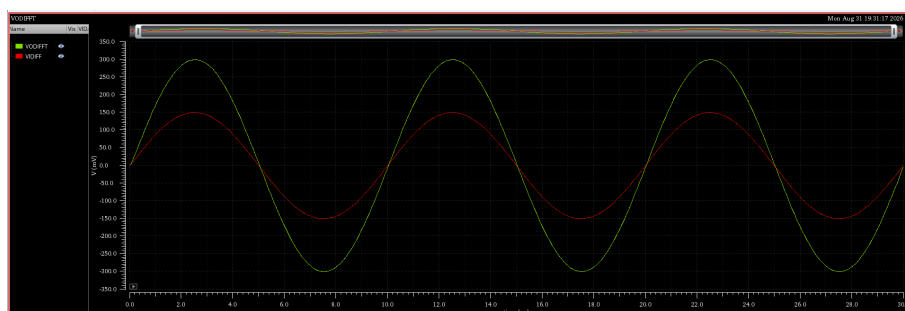


Figure 38: V_{IDIFF} and V_{ODIFF} overlaid

Swing Calculation

Peak-to-Peak Swings:

Diff Input Swing: 300 mV

Diff Output Swing: 600 mV

Calculated Closed Loop Gain: ≈ 2

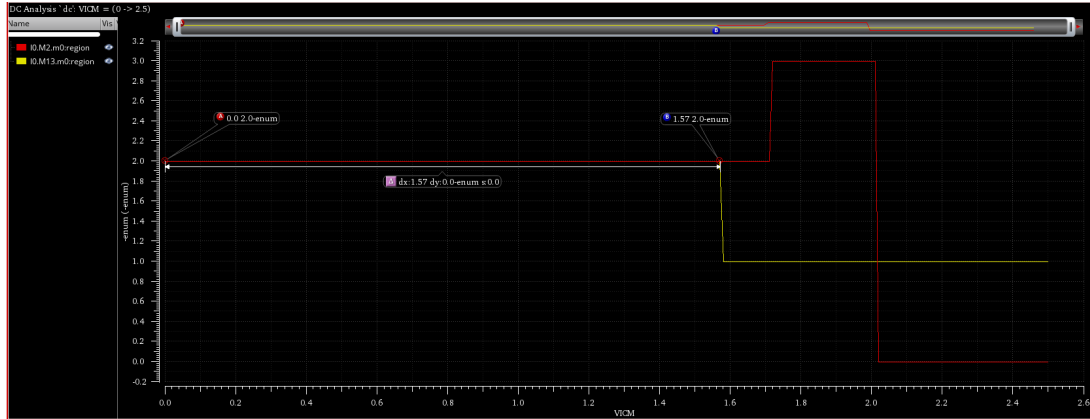


Figure 39: Input Common Mode Range (M2 $\rightarrow$ input pair & M13 $\rightarrow$ tail current source)

As shown in Figure 35, Settling Time was off the specification. To fix that, I decreased W (to $3.88 \mu m$) and L (to $320 nm$) of the Cascode transistors to decrease their C_{dd} . However, This procedure slightly consumed more current due to the mismatch in V_{DS} as L decreased and λ increased. Furthermore, To save power and area. I decreased the current consumption of the NMOS bias network to $2.5 \mu A$ and PMOS bias network to $2 \mu A$.

Test	Output	Nominal	Spec	Weight	Pass/Fail
FoldedCascode:TB_CL:1	ACL	1.998			
FoldedCascode:TB_CL:1	BWCL	9.034M			
FoldedCascode:TB_CL:1	GBWCL	18.05M			
FoldedCascode:TB_CL:1	VODIFF				
FoldedCascode:TB_CL:1	PM_DIFF	89.03			
FoldedCascode:TB_CL:1	Settletime	98.59n	< 100n		pass

Figure 40: Closed Loop Parameters

Test	Output	Nominal	Spec
FoldedCascode:TB_REAL:1	A0	3.517k	
FoldedCascode:TB_REAL:1	BW	15.52k	
FoldedCascode:TB_REAL:1	UGF	55.07M	
FoldedCascode:TB_REAL:1	GBW	54.59M	

Figure 41: Open Loop Parameters

Table 2: Transistor Sizing and Area Calculation

Circuit Block	Transistor(s)	W (μm)	L (μm)	Multiplier (m)	Area (μm^2)
Main OTA	M0	9.64	1.00	2	19.28
	M1	9.64	1.00	3	28.92
	M12	2.88	0.46	1	1.32
	M13	9.64	1.00	4	38.56
	M2, M3	12.44	0.31	2	15.43
	M4, M6	9.64	1.00	1	19.28
	M5, M7	3.88	0.32	1	2.4832
	M8, M10	1.56	0.40	1	1.25
	M9, M11	5.16	0.92	1	9.49
CMFB	M0, M1, M2, M3, M9	4.82	1.00	1	24.10
	M4, M5, M10	2.88	0.31	1	2.68
	M6, M8	1.56	0.80	1	2.50
	M7	1.72	0.92	1	1.58
Bias Network	M7	1.745	1.80	1	3.141
	M8	1.745	0.72	1	1.2564
	M4	0.345	0.46	1	0.1587
	M0	3.855	1.00	1	3.855
	M1	2.495	0.80	1	1.996
	M2	1.295	9.80	1	12.691
Total Area					189.9713

Power Calculation:Table 3: Power Consumption Breakdown by Circuit Block ($V_{DD} = 2.5\text{ V}$)

Circuit Block	Current (μA)	Power (μW)	% of Total
Main OTA	56.85	142.13	76.49%
Common-Mode Feedback (CMFB)	12.56	31.39	16.89%
Bias Network	4.92	12.31	6.62%
Total (from V_{DD})	74.33	185.82	100.00%

Figure of Merit (FoM):

$$\text{FoM} = \frac{GBW_{CL}}{I_{\text{total}} \times \text{Area}_{\text{total}}}$$

$$\text{FoM} = \frac{18.05\text{ MHz}}{69.41\text{ }\mu\text{A} \times 166.8732\text{ }\mu\text{m}^2}$$

$$\text{FoM} = \frac{18.05\text{ MHz}}{0.06941\text{ mA} \times 166.8732\text{ }\mu\text{m}^2}$$

$$\text{FoM} \approx 1.558 \frac{\text{MHz}}{\text{mA} \cdot \mu\text{m}^2}$$

Parameter	Target	Achieved
Closed loop gain	2	1.998
Phase margin	$\geq 70^\circ$	89.03°
CM input range – low	$\leq 0\text{ V}$	0 V
CM input range – high	$\geq 1\text{ V}$	1.57 V
Differential output swing	$1.2\text{ V}_{\text{pk-to-pk}}$	$1.19855\text{ V}_{\text{pk-to-pk}}$
DC Loop gain	60 dB	61.5663 dB
CL settling time for 1% error	100 ns	98.59 ns

Table 4: Target vs. Achieved Folded Cascode OTA Specifications